항목	내용	검토 상태
산출 유형	TEST_ONLY / SYNTHETIC_TEST_ONLY	실제 사업장 공정 결과가 아님
변경 제안 ID	861642b5-33fa-4734-9d96-216b082c3513	Append-only validation event 기반
변경 유형	PIPE_CHANGE	
공정 / 시트	SYNTHETIC-P / SYNTHETIC-01	
설비 관리번호	SYN-EQ-A	
계산 상태	CALCULATED_SYNTHETIC	시나리오 계산 요약
검토 상태	REVIEW_REQUIRED	
물리 연결 증빙	NOT_VERIFIED_NO_PHYSICAL_CONNECTION_EVIDENCE	증빙 없는 실제 연결은 미확정
실제 계산 readiness	CALCULATION_UNAVAILABLE	합성 계산은 실제 수지를 대체하지 않음
변경 전 네트워크	browser-before-pipe / CALCULATED	Synthetic before
변경 후 네트워크	browser-after-pipe / CALCULATED	Synthetic after
변경 전 설비 수		2 baseline result
변경 후 설비 수		2 baseline result
증빙 객체	source=SYNTHETIC_FIXTURE; status=UNVERIFIED; notice=No real-world evidence	입력 증빙은 자동 승인되지 않음
변경 전 snapshot	sourceProcess=SYNTHETIC-P; sourceEquipment=SYN-EQ-A; targetProcess=SYNTHETIC-P; targetEquipment=SYN-EQ-B; connectionId=SYN-A-B; connectionType=SERIES; materialClass=WASTEWATER; materialType=synthetic wastewater; unit=ton/day; period=2026-11-01/TEST_ONLY_BROWSER	제안 입력
변경 후 snapshot	sourceProcess=SYNTHETIC-P; sourceEquipment=SYN-EQ-A; targetProcess=SYNTHETIC-P; targetEquipment=SYN-EQ-C; connectionId=SYN-A-C; connectionType=SERIES; materialClass=WASTEWATER; materialType=synthetic wastewater; unit=ton/day; period=2026-11-01/TEST_ONLY_BROWSER	제안 입력

구분	시나리오	연결 ID	출발 공정	출발 설비/경계	도착 공정	도착 설비/경계	물질 분류	물질 종류	연결 유형	유량	단위
변경 전	BASELINE	A-OUT	SYNTHETIC-P	SYN-EQ-A		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		80 ton/day
변경 전	BASELINE	B-OUT	SYNTHETIC-P	SYN-EQ-B		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		20 ton/day
변경 전	BASELINE	IN-A		EXTERNAL:IN	SYNTHETIC-P	SYN-EQ-A	WASTEWATER	synthetic wastewater	EXTERNAL_INFLOW		100 ton/day
변경 전	BASELINE	SYN-A-B	SYNTHETIC-P	SYN-EQ-A	SYNTHETIC-P	SYN-EQ-B	WASTEWATER	synthetic wastewater	SERIES		20 ton/day
변경 전	FLOW-10	A-OUT	SYNTHETIC-P	SYN-EQ-A		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		90 ton/day
변경 전	FLOW-10	B-OUT	SYNTHETIC-P	SYN-EQ-B		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		10 ton/day
변경 전	FLOW-10	IN-A		EXTERNAL:IN	SYNTHETIC-P	SYN-EQ-A	WASTEWATER	synthetic wastewater	EXTERNAL_INFLOW		100 ton/day
변경 전	FLOW-10	SYN-A-B	SYNTHETIC-P	SYN-EQ-A	SYNTHETIC-P	SYN-EQ-B	WASTEWATER	synthetic wastewater	SERIES		10 ton/day
변경 전	FLOW-50	A-OUT	SYNTHETIC-P	SYN-EQ-A		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		50 ton/day
변경 전	FLOW-50	B-OUT	SYNTHETIC-P	SYN-EQ-B		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		50 ton/day
변경 전	FLOW-50	IN-A		EXTERNAL:IN	SYNTHETIC-P	SYN-EQ-A	WASTEWATER	synthetic wastewater	EXTERNAL_INFLOW		100 ton/day
변경 전	FLOW-50	SYN-A-B	SYNTHETIC-P	SYN-EQ-A	SYNTHETIC-P	SYN-EQ-B	WASTEWATER	synthetic wastewater	SERIES		50 ton/day
변경 후	BASELINE	A-OUT	SYNTHETIC-P	SYN-EQ-A		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		80 ton/day
변경 후	BASELINE	C-OUT	SYNTHETIC-P	SYN-EQ-C		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		20 ton/day
변경 후	BASELINE	IN-A		EXTERNAL:IN	SYNTHETIC-P	SYN-EQ-A	WASTEWATER	synthetic wastewater	EXTERNAL_INFLOW		100 ton/day
변경 후	BASELINE	SYN-A-C	SYNTHETIC-P	SYN-EQ-A	SYNTHETIC-P	SYN-EQ-C	WASTEWATER	synthetic wastewater	SERIES		20 ton/day
변경 후	FLOW-10	A-OUT	SYNTHETIC-P	SYN-EQ-A		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		90 ton/day
변경 후	FLOW-10	C-OUT	SYNTHETIC-P	SYN-EQ-C		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		10 ton/day
변경 후	FLOW-10	IN-A		EXTERNAL:IN	SYNTHETIC-P	SYN-EQ-A	WASTEWATER	synthetic wastewater	EXTERNAL_INFLOW		100 ton/day
변경 후	FLOW-10	SYN-A-C	SYNTHETIC-P	SYN-EQ-A	SYNTHETIC-P	SYN-EQ-C	WASTEWATER	synthetic wastewater	SERIES		10 ton/day
변경 후	FLOW-50	A-OUT	SYNTHETIC-P	SYN-EQ-A		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		50 ton/day
변경 후	FLOW-50	C-OUT	SYNTHETIC-P	SYN-EQ-C		EXTERNAL:OUT	WASTEWATER	synthetic wastewater	EXTERNAL_OUTFLOW		50 ton/day
변경 후	FLOW-50	IN-A		EXTERNAL:IN	SYNTHETIC-P	SYN-EQ-A	WASTEWATER	synthetic wastewater	EXTERNAL_INFLOW		100 ton/day
변경 후	FLOW-50	SYN-A-C	SYNTHETIC-P	SYN-EQ-A	SYNTHETIC-P	SYN-EQ-C	WASTEWATER	synthetic wastewater	SERIES		50 ton/day

구분	시나리오	연결 ID	기간	계산 상태	제약	유량 근거	고정	증빙
변경 전	BASELINE	A-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	BASELINE	B-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	BASELINE	IN-A	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	BASELINE	SYN-A-B	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	FLOW-10	A-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	FLOW-10	B-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	FLOW-10	IN-A	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	FLOW-10	SYN-A-B	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	FLOW-50	A-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	FLOW-50	B-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	FLOW-50	IN-A	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 전	FLOW-50	SYN-A-B	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	BASELINE	A-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	BASELINE	C-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	BASELINE	IN-A	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	BASELINE	SYN-A-C	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	FLOW-10	A-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	FLOW-10	C-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	FLOW-10	IN-A	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	FLOW-10	SYN-A-C	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	FLOW-50	A-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	FLOW-50	C-OUT	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		CALCULATED	false	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	FLOW-50	IN-A	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified
변경 후	FLOW-50	SYN-A-C	2026-11-01/ TEST_ONLY_BROWSER	CALCULATED		USER_INPUT	true	USER-SUPPLIED TEST_ONLY: Synthetic TEST_ONLY fixture; actual route unverified

비교 상태	연결 ID	변경 전 출발	변경 전 도착	변경 후 출발	변경 후 도착	변경 전 유량	변경 후 유량	변경 전 단위	변경 후 단위
DELETED	B-OUT	SYNTHETIC-P / SYN-EQ-B	외부 / EXTERNAL:OUT				20	ton/day	
ADDED	C-OUT			SYNTHETIC-P / SYN-EQ-C	외부 / EXTERNAL:OUT			20	ton/day
DELETED	SYN-A-B	SYNTHETIC-P / SYN-EQ-A	SYNTHETIC-P / SYN-EQ-B				20	ton/day	
ADDED	SYN-A-C			SYNTHETIC-P / SYN-EQ-A	SYNTHETIC-P / SYN-EQ-C			20	ton/day

비교 상태	연결 ID	변경 전 출발	변경 전 유형	변경 후 유형	변경 전 물질	변경 후 물질	변경 전 기간	변경 후 기간
DELETED	B-OUT	SYNTHETIC-P / SYN-EQ-B	EXTERNAL_OUTFLOW		WASTEWATER / synthetic wastewater		2026-11-01/ TEST_ONLY_BROWSER	
ADDED	C-OUT			EXTERNAL_OUTFLOW		WASTEWATER / synthetic wastewater		2026-11-01/ TEST_ONLY_BROWSER
DELETED	SYN-A-B	SYNTHETIC-P / SYN-EQ-A	SERIES		WASTEWATER / synthetic wastewater		2026-11-01/ TEST_ONLY_BROWSER	
ADDED	SYN-A-C			SERIES		WASTEWATER / synthetic wastewater		2026-11-01/ TEST_ONLY_BROWSER

구분	시나리오	공정	물질 분류	물질 종류	단위	유입	유출	발생/소모	잔차
변경 전	BASELINE	SYNTHETIC-P	WASTEWATER	synthetic wastewater	ton/day	100	100	0	0
변경 전	FLOW-10	SYNTHETIC-P	WASTEWATER	synthetic wastewater	ton/day	100	100	0	0
변경 전	FLOW-50	SYNTHETIC-P	WASTEWATER	synthetic wastewater	ton/day	100	100	0	0
변경 후	BASELINE	SYNTHETIC-P	WASTEWATER	synthetic wastewater	ton/day	100	100	0	0
변경 후	FLOW-10	SYNTHETIC-P	WASTEWATER	synthetic wastewater	ton/day	100	100	0	0
변경 후	FLOW-50	SYNTHETIC-P	WASTEWATER	synthetic wastewater	ton/day	100	100	0	0

구분	시나리오	공정	상태	측정 기간
변경 전	BASELINE	SYNTHETIC-P	BALANCED	2026-11-01/test_only_browser
변경 전	FLOW-10	SYNTHETIC-P	BALANCED	2026-11-01/test_only_browser
변경 전	FLOW-50	SYNTHETIC-P	BALANCED	2026-11-01/test_only_browser
변경 후	BASELINE	SYNTHETIC-P	BALANCED	2026-11-01/test_only_browser
변경 후	FLOW-10	SYNTHETIC-P	BALANCED	2026-11-01/test_only_browser
변경 후	FLOW-50	SYNTHETIC-P	BALANCED	2026-11-01/test_only_browser

공정	물질 종류	단위	측정 기간	변경 전 잔차	변경 후 잔차	잔차 delta	비교 상태
SYNTHETIC-P	synthetic wastewater	ton/day	2026-11-01/test_only_browser	0	0	0	SYNTHETIC_COMPARISON

구분	시나리오	물질 분류	물질 종류	단위	외부 유입 합계	외부 유출 합계	잔차	상태
변경 전	BASELINE	WASTEWATER	synthetic wastewater	ton/day	100	100	0	BALANCED
변경 전	FLOW-10	WASTEWATER	synthetic wastewater	ton/day	100	100	0	BALANCED
변경 전	FLOW-50	WASTEWATER	synthetic wastewater	ton/day	100	100	0	BALANCED
변경 후	BASELINE	WASTEWATER	synthetic wastewater	ton/day	100	100	0	BALANCED
변경 후	FLOW-10	WASTEWATER	synthetic wastewater	ton/day	100	100	0	BALANCED
변경 후	FLOW-50	WASTEWATER	synthetic wastewater	ton/day	100	100	0	BALANCED

구분	시나리오	물질 분류	기간
변경 전	BASELINE	WASTEWATER	2026-11-01/test_only_browser
변경 전	FLOW-10	WASTEWATER	2026-11-01/test_only_browser
변경 전	FLOW-50	WASTEWATER	2026-11-01/test_only_browser
변경 후	BASELINE	WASTEWATER	2026-11-01/test_only_browser
변경 후	FLOW-10	WASTEWATER	2026-11-01/test_only_browser
변경 후	FLOW-50	WASTEWATER	2026-11-01/test_only_browser

비교 상태	물질 분류	물질 종류	단위	기간	변경 전 외부 유입	변경 후 외부 유입	변경 전 외부 유출	변경 후 외부 유출	
SYNTHETIC_COMPARISON	WASTEWATER	synthetic wastewater	ton/day	2026-11-01/test_only_browser		100	100	100	100

비교 상태	물질 분류	물질 종류	변경 전 잔차	변경 후 잔차
SYNTHETIC_COMPARISON	WASTEWATER	synthetic wastewater	0	0

구분	항목/상태	상세
변경 전 / FLOW-10	영향 공정	SYNTHETIC-P
변경 전 / FLOW-10	영향 설비	SYN-EQ-A, SYN-EQ-B
변경 전 / FLOW-50	영향 공정	SYNTHETIC-P
변경 전 / FLOW-50	영향 설비	SYN-EQ-A, SYN-EQ-B
변경 후 / FLOW-10	영향 공정	SYNTHETIC-P
변경 후 / FLOW-10	영향 설비	SYN-EQ-A, SYN-EQ-C
변경 후 / FLOW-50	영향 공정	SYNTHETIC-P
변경 후 / FLOW-50	영향 설비	SYN-EQ-A, SYN-EQ-C
검토 및 제약	상태	상세
합성 데이터	TEST_ONLY	입력 네트워크는 물리 연결 증빙이 아님
실제 공정 readiness	CALCULATION_UNAVAILABLE	실제 연결 검증이 완료되지 않음
연결 증빙	NOT_VERIFIED_NO_PHYSICAL_CONNECTION_EVIDENCE	source=SYNTHETIC_FIXTURE; status=UNVERIFIED; notice=No real-world evidence
변경 전	baseline 계산 상태	CALCULATED
변경 전 / FLOW-10	시나리오 계산 상태	CALCULATED
변경 전 / FLOW-50	시나리오 계산 상태	CALCULATED
변경 후	baseline 계산 상태	CALCULATED
변경 후 / FLOW-10	시나리오 계산 상태	CALCULATED
변경 후 / FLOW-50	시나리오 계산 상태	CALCULATED